*"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."*

**Roll No.** 24UECC8053

**B.E. III (ECC)**

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**DOS**

# THIRD B.E. EXAMINATION - 2026
# ELECTRONICS & COMPUTER ENGINEERING
## (VI SEMESTER CBCS)
## (For Main/ATKT/Ex Students)
## 6ECC43A(ECE): DESIGN OF OPERATING SYSTEMS (ECC)

Time - Three Hours

Maximum Marks - 70

Note:- (i) Attempt any **Five** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.

1. (a) Explain Multiprograming in operating system and list its advantages & disadvantages. 7

   (b) Define an operating system and explain its functions. 7

2. (a) Explain the components of a process and the role of Process Control Block (PBC). 7

(b) Define Process Scheduling and explain types of Process Schedulers. 7

3. (a) A system uses Round Robin scheduling with time quantum = 4ms. Processes arrive at time O with burst time:

| Process | Burst Time |
|---|---|
| $P_1$ | 10 ms |
| $P_2$ | 5 ms |
| $P_3$ | 8 ms |

Draw the Gantt chart and calculate the average turnaround time and average waiting time. 7

(b) Explain the First come first serve (FCFS) CPU scheduling algorithm with suitable example. 7

4. (a) Explain Semaphores operations and types of semaphore. 7

(b) Define the Deadlock. How to prevent from deadlock? 7

5. (a) Explain Paging in memory management with diagram. 7

(b) Explain the optimal page replacement algorithm with an example. 7

6. (a) Explain Disk scheduling performance parameters. 7

(b) Explain SCAN disk scheduling algorithm with example. 7

7. (a) Explain monitors in process synchronization. 7

(b) What is a File System? Explain the basic file operations. 7